

AWS Containers Overview

Containers are lightweight, portable units that package applications and their dependencies. AWS provides multiple services to run and manage containers efficiently.

Key AWS Container Services

1. Amazon Elastic Container Service (ECS)

- Fully managed container orchestration service.
- Supports Docker containers.
- Integrates with AWS services like IAM, CloudWatch, and ALB.

2. AWS Fargate

- Serverless compute engine for containers.
- Works with ECS and EKS.
- No need to manage EC2 instances.

3. Amazon Elastic Container Registry (ECR)

- Secure, scalable container image registry.
- Supports private repositories and image scanning.

Use Case: Microservices, batch processing, containerized workloads.

AWS Kubernetes (Amazon EKS)

Amazon Elastic Kubernetes Service (EKS) is a managed Kubernetes service that simplifies running Kubernetes on AWS without managing control planes.

Key Features of EKS

- **Managed Control Plane**
AWS runs and scales the Kubernetes control plane.
- **Integration with AWS Services**
IAM for authentication, VPC for networking, CloudWatch for monitoring.
- **Supports Hybrid and Multi-Cloud**
Works with on-premises Kubernetes clusters using EKS Anywhere.

Components

- **Worker Nodes:** EC2 or Fargate instances running pods.

- **Control Plane: Managed by AWS.**
- **Networking: Uses VPC CNI plugin for pod networking.**

Use Case: Large-scale container orchestration, CI/CD pipelines, hybrid cloud deployments.